Lab 6 Specifications

Lab 6 asks you to build a complete IoT device on top of a driver you write yourself against the DS1722 register map. The main focus of your driver implementation is enabling SPI communication, and understanding of serial communication is a major learning goal for the lab.

Hardware Specifications

Proficiency

- ☐ DS1722 temperature sensor is correctly wired per its datasheet pinout.
- ☐ ESP8266 module is correctly wired and powered at 3.3V, with TX/RX crossed between the MCU and the ESP8266.
- ☐ FPGA is disconnected (DIP switches open) so it cannot interfere with pins shared with the MCU.
- ☐ System correctly reads the current temperature from the DS1722 sensor and displays it on a webpage that updates the temperature when refreshed.
- ☐ Webpage displays the current LED state and can toggle it.

Excellence

- ☐ User can select the DS1722's resolution (e.g., 8/9/10/11/12-bit) from the webpage, and the driver correctly reconfigures the sensor in response.

Software and Microcontroller Design Specifications

Proficiency

- ☐ SPI device driver is written using the CMSIS device templates (register structs and base addresses from `stm32l432xx.h`), not a third-party, copied driver or home-grown driver.

Excellence

- ☐ System resolution is configured such that it could correctly read temperatures across the sensor's -10°C to 30°C range.

Other Documentation

Proficiency

- ☐ Report includes a labeled logic analyzer capture of a full SPI read transaction (CE, SCK, SDO and SDI), as required in the lab writeup.

Excellence

- ☐ Logic analyzer capture is annotated to show how the observed timing matches the datasheet's specified SPI timing.

Schematic Specifications

Proficiency

- ☐ Schematic correctly represents the circuit that was built on the breadboard.
- ☐ No parts of motherboard schematic replicated. Shows only off-board components. Indicate on-board components with a brief paragraph that links the motherboard schematic (here).
- ☐ No parts internal to the chip (e.g., internal pullup resistors) drawn on schematic. Indicate important pullups, capacitors, etc. with comments instead.
- ☐ All pin names labeled.
- ☐ All pin numbers labeled.
- ☐ Crossing wires clearly identified as junction or unconnected.
- ☐ Neat layout (e.g., clear organization and spacing).
- ☐ All parts labeled with part number.
- ☐ All component values present.

Excellence

- ☐ Standard symbols used for all components where applicable.
- ☐ Signals "flow" from left to right where possible (e.g., inputs on left hand side, outputs on right hand side, power on top, ground on bottom)
- ☐ Title block with author name, title, and date.

Lab Writeup

Proficiency and Excellence

- ☐ Documentation embedded in a well-structured, concise report with one section per major deliverable. Sections should match structure of other specs.

- ☐ Spoiler tags are used to hide screenshots and code. Some unspoilered entries (~1 waveform, essential code snippets) allowed if they are essential to the narrative, kept to a minimum.
- ☐ Writeup contains minimal spelling or grammar issues and any errors do not significantly detract from clarity of the writeup.
- ☐ Statement of whether the design meets all the requirements. If not, list the shortcomings.
- ☐ AI prototype attempted and some reflection is recorded.
- ☐ Number of hours spent working on the lab are included.
- ☐ (Optional) List comments or suggestions on what was particularly good about the assignment or what you think needs to change in future versions.